SBT-DF203 – LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

Digital Forensics Investigation Report

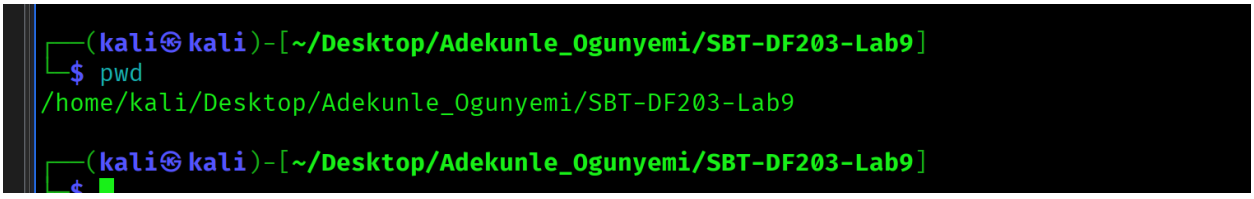
Course Information

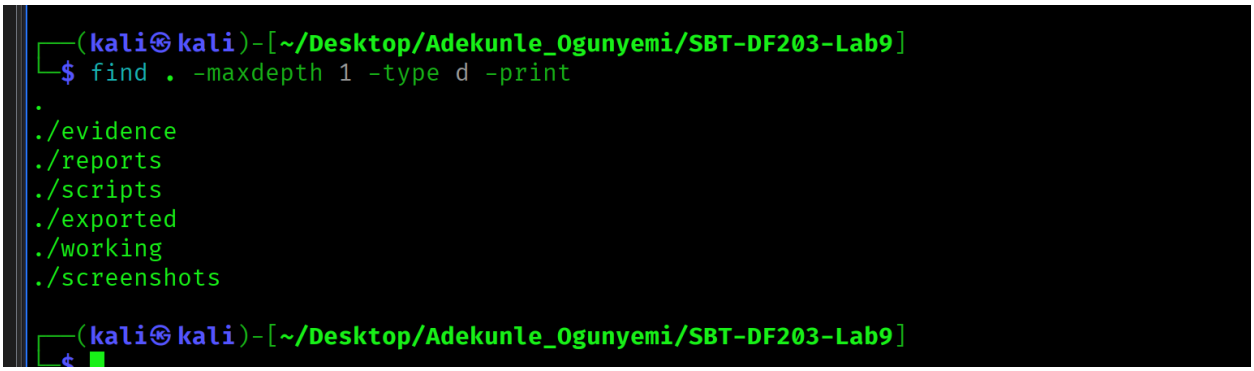
Field	Details
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	9
Lab Title	WEP40 Wireless Packet Decryption and Aircrack Forensics
Platform	Kali Linux Virtual Machine
Student	Adekunle Ogunyemi
Student ID	2025/FWSD/11479
Instructor	Aminu Idris, AMCPN
Date	25 September 2026

1. Executive Summary

This lab focused on the forensic analysis of a historical wireless packet capture that was protected using WEP40 encryption. The evidence was a compressed CodeGate CTF 2015 capture file, which I downloaded, preserved, decompressed, and analysed entirely offline inside an isolated Kali Linux VM. The main goal was to identify IEEE 802.11 traffic, validate the WEP40 key using Aircrack-ng, decrypt the capture with airdecap-ng, and then extract useful forensic artefacts such as IP/MAC endpoints, TCP conversations, images, and HTML files.

The capture contained 45,169 packets, of which 15,713 were data frames, and 15,477 were WEP-protected data packets. Aircrack-ng successfully recovered the 40-bit key A4:3D:F6:F3:74 using 15,477 IVs. The key was entered into airdecap-ng without colons (A43DF6F374), which decrypted 100% of the WEP data packets. After decryption, higher-layer protocols such as ARP, DHCP, DNS, TCP, TLSv1, and IGMPv2 became visible. The main internal host was 192.168.0.15 (Apple_68:96:7c), and the main external server was 198.90.20.111 on port 80. Several images, HTML files, CSS files, JavaScript files, and font files were carved from the decrypted traffic. The lab confirmed that WEP is cryptographically broken and should be replaced with WPA2-AES/CCMP or WPA3.





2. Authority, Scope and Evidence Description

2.1 Authority

This lab was completed as part of the SBT-DF203 course at the International Cybersecurity and Digital Forensics Academy (ICDFA). All work was performed inside an authorised Kali Linux virtual machine. The capture was a supplied historical training file from the CodeGate CTF 2015 Good Crypto challenge. No live wireless networks were targeted, and no third-party systems were accessed.

2.2 Scope

The scope of this lab included:

- Preserving and hashing the supplied file.xz evidence
- Decompressing the capture and verifying its file type
- Identifying 802.11 management, control, and data frames
- Confirming WEP protection and extracting IV-related fields
- Recovering or validating the WEP40 key with Aircrack-ng
- Decrypting the capture offline with airdecap-ng
- Extracting IP/MAC endpoints and TCP conversations
- Carving images, HTML, and other objects from the decrypted traffic
- Documenting the security weaknesses of WEP

2.3 Evidence Description

Evidence Item	Description	Size
evidence/file.xz	Original compressed CodeGate CTF capture	12,837,660 bytes
working/file_working	Decompressed pcap capture	14,234,002 bytes
working/file_working-dec	Decrypted capture produced by airdecap-ng	12,784,501 bytes

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ sudo wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programmi
ng/good-crypto/file.xz'
--2026-09-26 06:49:30--  https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codeg
ate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199
.109.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443 ...
 connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: ‘evidence/file.xz’

evidence/file.xz       100%[═══════════════════════════⟹]  12.24M  3.01MB/s    in 4.4s

2026-09-26 06:49:36 (2.81 MB/s) - ‘evidence/file.xz’ saved [12837660/12837660]
```

3.1 Working Environment

- Operating System: Kali Linux VM
- Working Directory: /home/kali/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9
- Analysis Date: 26 September 2026
- Tools Used: Aircrack-ng, airdecap-ng, Wireshark, tshark, foremost, binwalk, file, strings, xz-utils, wget
- Network Mode: Offline analysis only.

3.2 Folder Structure

I created the required folder structure before downloading the evidence:

mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ find . -maxdepth 1 -type d -print
.
./evidence
./reports
./scripts
./exported
./working
./screenshots

┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$
```

3.3 Evidence Acquisition

The evidence file was downloaded from the supplied GitHub URL:
sudo wget -O evidence/file.xz \
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'

The download completed successfully (12.24 MB in 4.4 seconds).

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ sudo wget -O evidence/file.xz \
  'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programmi
ng/good-crypto/file.xz'
--2026-09-26 06:49:30--  https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codeg
ate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199
.109.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443 ...
 connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: ‘evidence/file.xz’

evidence/file.xz       100%[═══════════════════════════════>]  12.24M  3.01MB/s    in 4.4s

2026-09-26 06:49:36 (2.81 MB/s) - ‘evidence/file.xz’ saved [12837660/12837660]
```

4. Evidence Integrity and Chain of Custody

4.1 Chain of Custody Worksheet

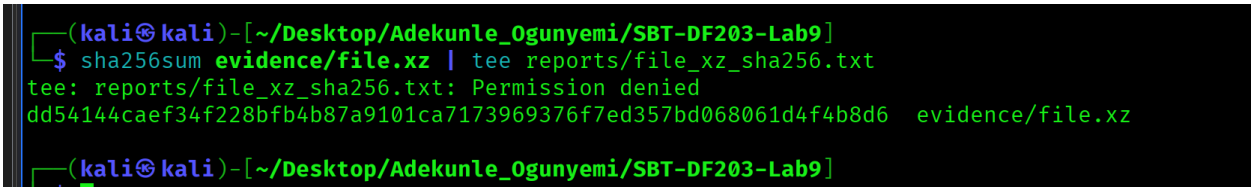
Field	Entry
Case/Lab Identifier	SBT-DF203-Lab9-AdekunleOgunyemi
Trainee Name	Adekunle Ogunyemi
ID Number	2025/FWSD/11479
Date/Time Started	26 September 2026 06:49 UTC
Evidence File Name	file.xz
Source/Method	Downloaded from CodeGate CTF 2015 GitHub repository
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6
Working-copy SHA-256	Same as original (verified)
Analysis VM	Kali Linux
Notes on Changes	None – original preserved; working copy used for analysis

4.2 Hash Verification

The original compressed file was hashed:
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt

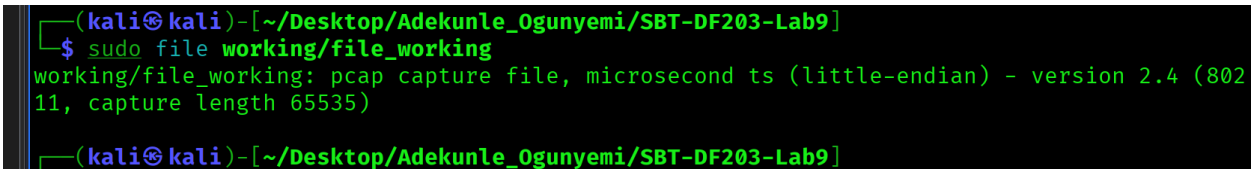
Output:

dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz



The file was then copied to the working directory and decompressed:
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working

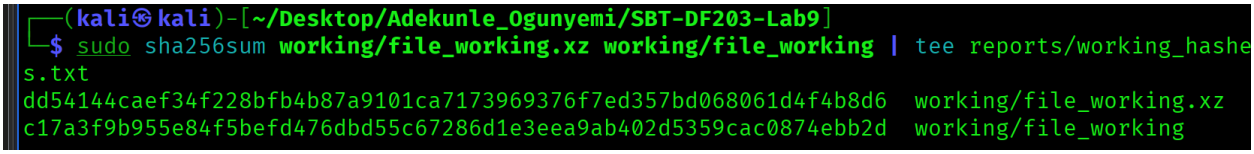
Output:
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)



The working files were hashed:

sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt

Output:
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working



The original and working copy hashes matched, confirming that the evidence was preserved without alteration.

5. Part A – Inventory the Wireless Capture

5.1 Capture Metadata

The capture was analysed with capinfos:

capinfos "$CAP" | tee reports/capinfos.txt

Output:

- File type: Wireshark/tcpdump pcap
- Encapsulation: IEEE 802.11 Wireless LAN
- Number of packets: 45,169
- File size: 14 MB
- Capture duration: 272.969795 seconds
- Earliest packet: 2015-03-06 02:51:54.364525
- Latest packet: 2015-03-06 02:56:27.334320
- Data bit rate: 395 kbps
- Average packet size: 299.13 bytes
- SHA256: c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ capinfos "$CAP" | tee reports/capinfos.txt
File name:           working/file_working
File type:           Wireshark/tcpdump/ ... - pcap
File encapsulation:  IEEE 802.11 Wireless LAN
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 65535 bytes
Number of packets:   45 k
File size:           14 MB
Data size:           13 MB
Capture duration:    272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:   2015-03-06 02:56:27.334320
Data byte rate:      49 kBps
Data bit rate:       395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:              c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:                cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:   False
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
                     Capture length = 65535
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
                     Number of packets = 45169

┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$
```

5.2 Protocol Hierarchy

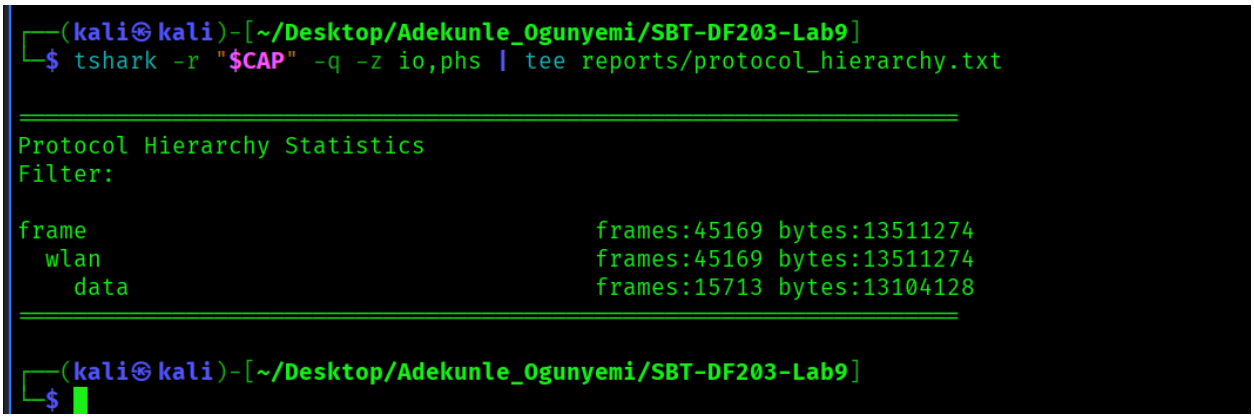
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

Output showed:

- frame: 45,169 frames / 13,511,274 bytes

- wlan: 45,169 frames / 13,511,274 bytes
- data: 15,713 frames / 13,104,128 bytes

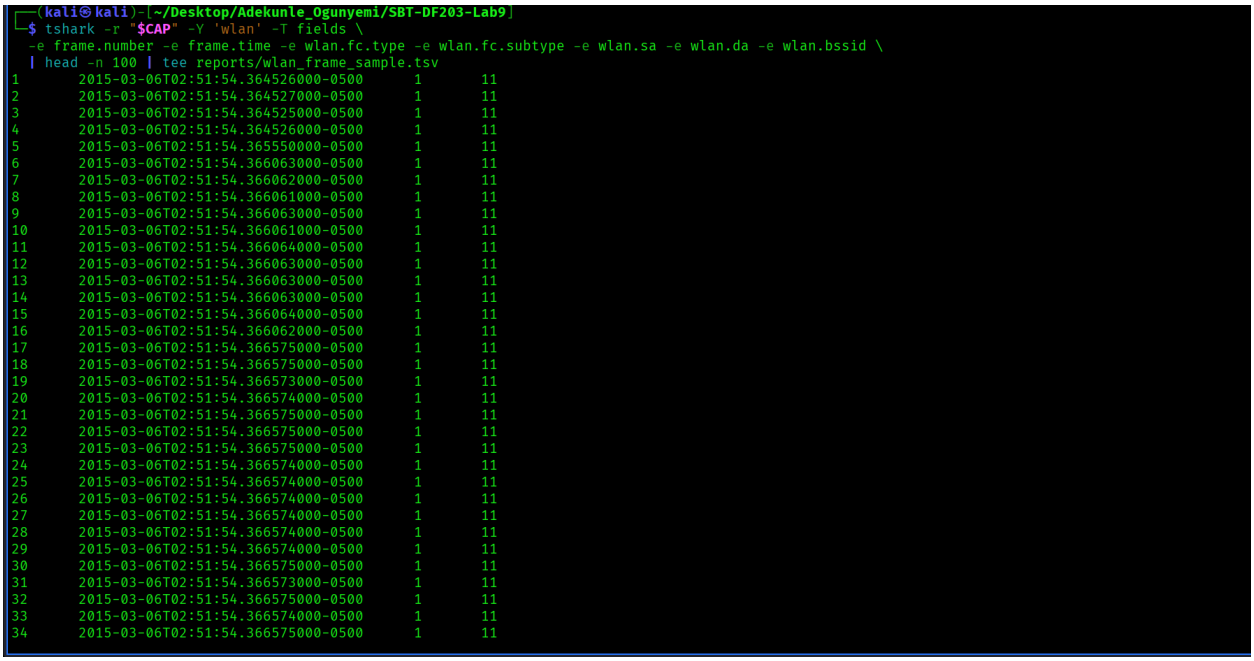
This confirmed that the capture is entirely 802.11 wireless traffic, with a large portion being data frames.



5.3 802.11 Frame Categories

tshark -r "$CAP" -Y 'wlan' -T fields \
-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv

The sample showed type 1 frames (control frames) with subtype 11. This means the first 100 frames were control frames used for flow control and acknowledgements.



Frame Category	Type Value	Purpose	Example Evidence
Management	0	Association, authentication, beacons	Not dominant in sample
Control	1	Flow control and acknowledgements	Type 1, subtype 11
Data	2	Carries network payload	15,713 data frames

6. Part B – Identify WEP Protection and Key Parameters

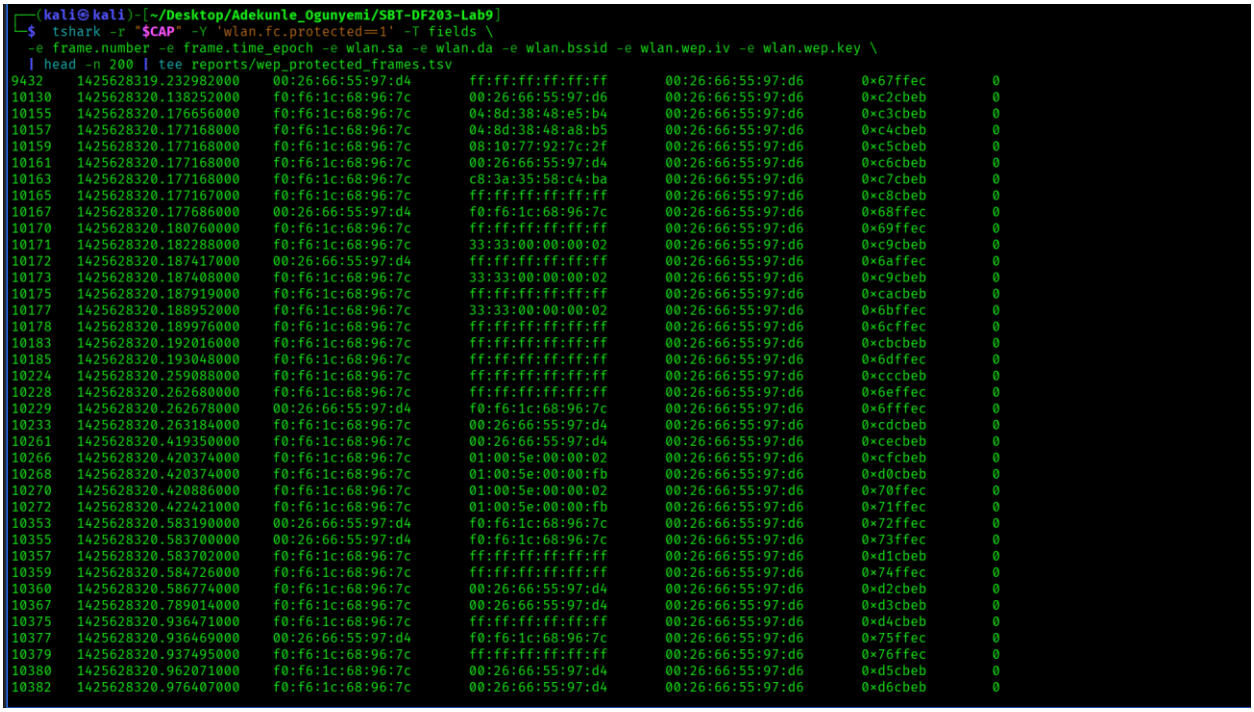
6.1 Protected Frames

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv

The output confirmed that protected data frames were present. Key observations:

- BSSID: 00:26:66:55:97:d6

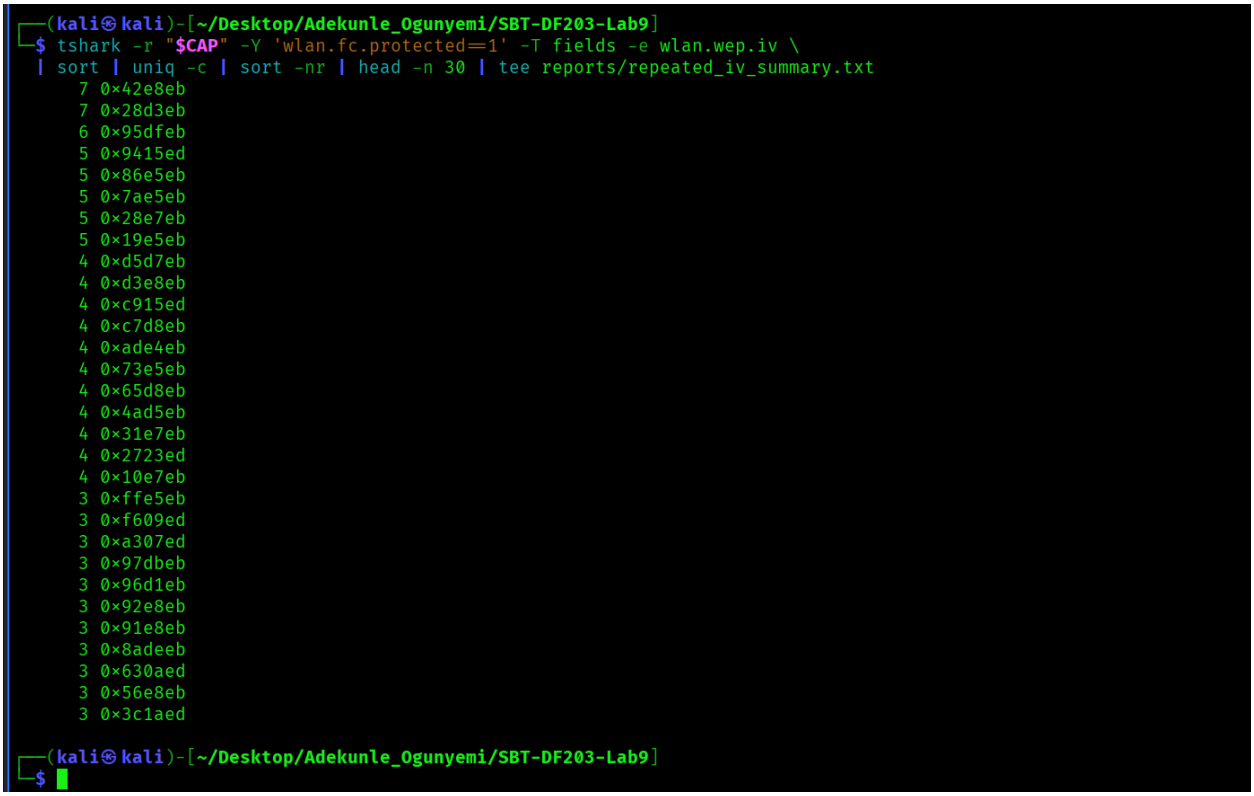
- Source MAC examples: 00:26:66:55:97:d4, 04:8d:38:48:8e:b4, 08:10:77:92:7c:f2, c8:3a:55:97:d6
- Destination MAC examples: ff:ff:ff:ff:ff:ff, 00:26:66:55:97:d6
- IV values: 0x6f7fcc, 0x2cbcbb, 0x3cbb, 0x4cbb, 0x5cbb, 0x6c6bb, 0x8cbb, 0x6effc, 0x69ffc, 0x9cbb



6.2 Repeated IV Summary

tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 | tee
reports/repeated_iv_summary.txt

The output showed that many IVs appeared repeatedly, which is a known weakness of WEP. Repeated IVs mean the same RC4 keystream is reused, which leaks relationships between plaintexts and enables statistical key recovery.



6.3 WEP40 Composition

WEP40, also called 64-bit WEP, combines a 24-bit Initialisation Vector (IV) with a 40-bit secret key. The IV is transmitted in plaintext with each frame. Because the IV space is only 2^24, busy networks reuse IVs quickly. RC4 also has known keystream biases. Together, these weaknesses make WEP unsuitable for protecting wireless traffic.

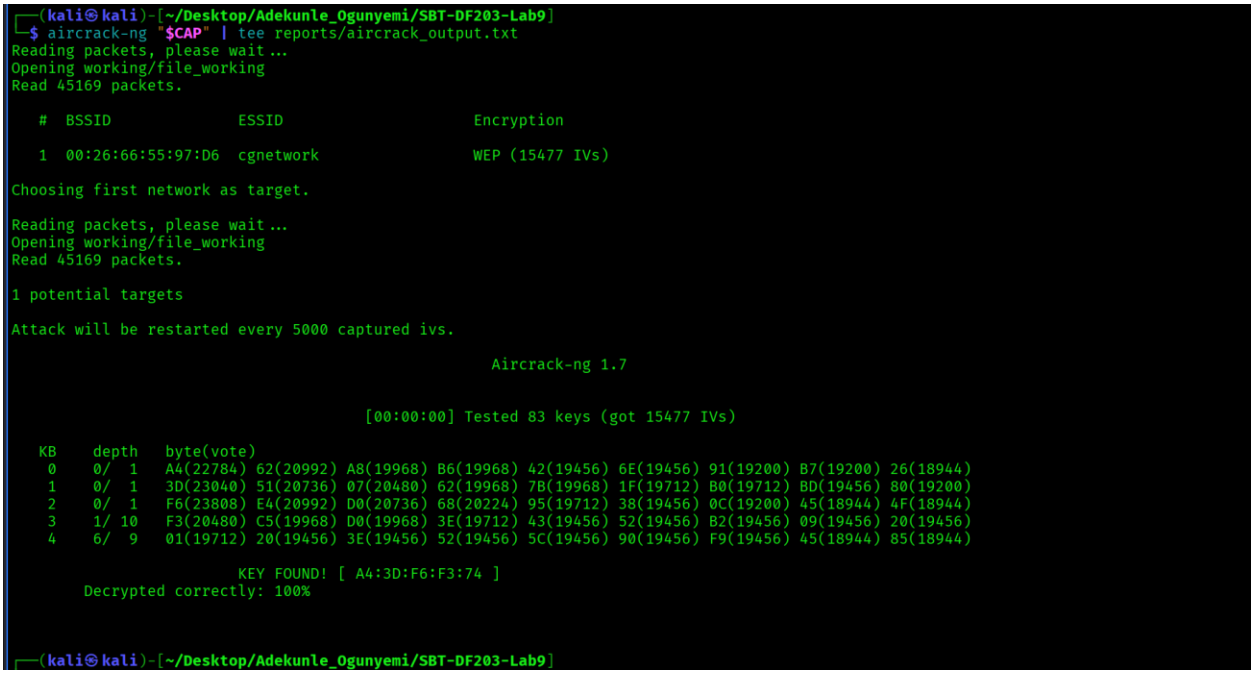
7. Part C – Recover or Validate the WEP40 Key

7.1 Aircrack-ng Execution

aircrack-ng "$CAP" | tee reports/aircrack_output.txt

Output:
1 0:26:66:55:97:D6 cgnetwork WEP (15477 IVs)
Choosing first network as target.
Reading packets, please wait...

Opening working/file_working
Read 45169 packets.
1 potential targets
Attack will be restarted every 5000 captured ivs.
Aircrack-ng 1.7
[00:00:00] Tested 83 keys (got 15477 IVs)
KEY FOUND! [A4:3D:F6:F3:74]
Decrypted correctly: 100%

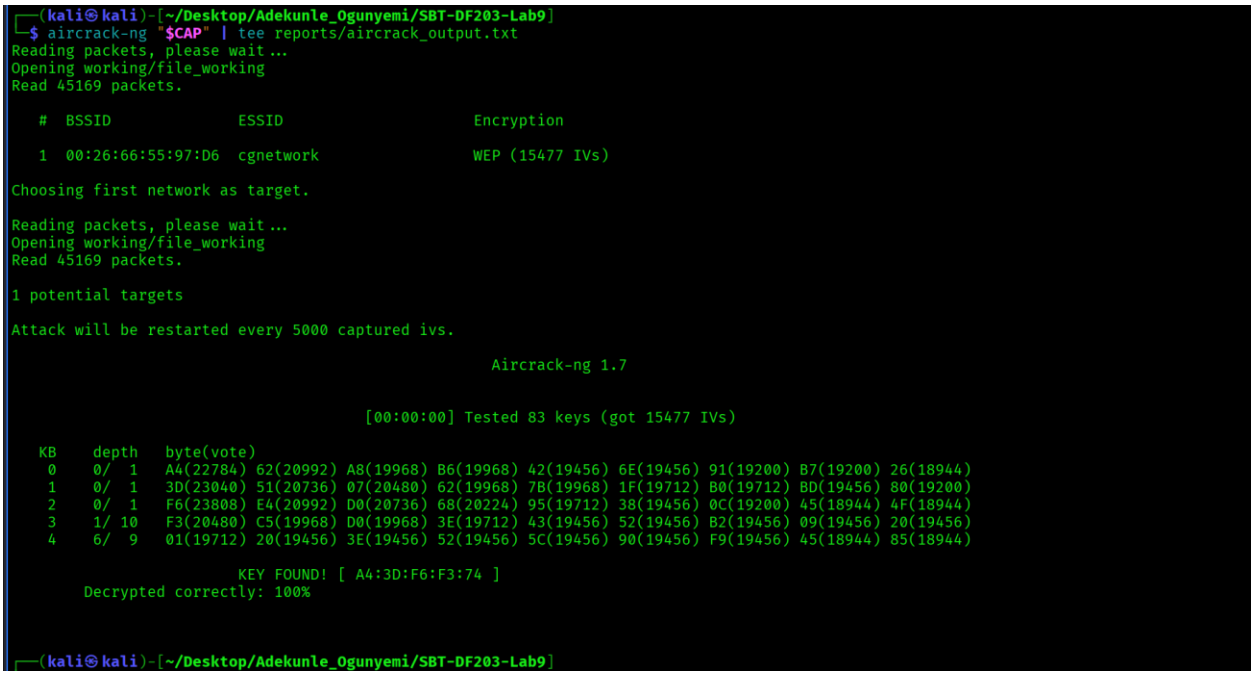


7.2 Key Validation

The recovered key was A4:3D:F6:F3:74. This matches the historical training key provided in the slide deck. The key was documented in the reports folder:

printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt

This key is part of the historical CTF dataset and is not authorization to attack any wireless network.



8. Part D – Decrypt the Capture Offline

8.1 Airdecap-ng Decryption

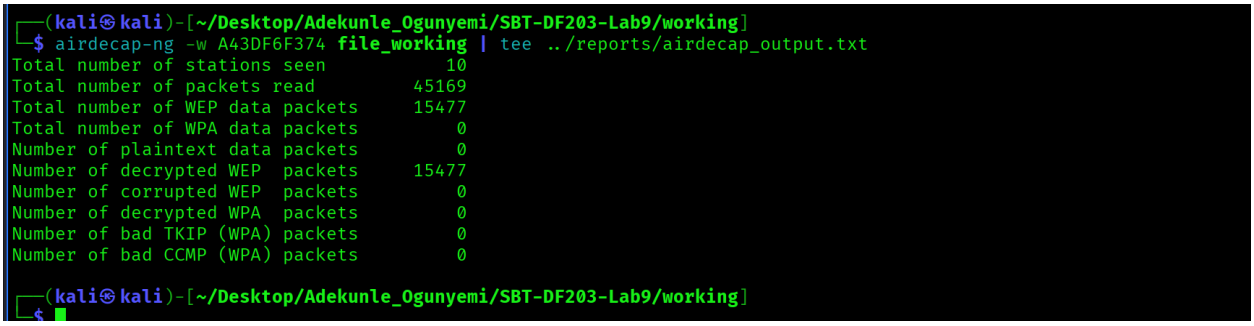
cd working
airdecap-ng -w A43DF6F374 file_working | tee
../reports/airdecap_output.txt
cd ..

Output:

Total number of stations seen 10
Total number of packets read 45169
Total number of WEP data packets 15477
Total number of WPA data packets 0
Number of plaintext data packets 0
Number of decrypted WEP packets 15477
Number of corrupted WEP packets 0
Number of decrypted WPA packets 0
Number of bad TKIP (WPA) packets 0

Number of bad CCMP (WPA) packets 0

All 15,477 WEP data packets were decrypted successfully with zero corrupted packets.

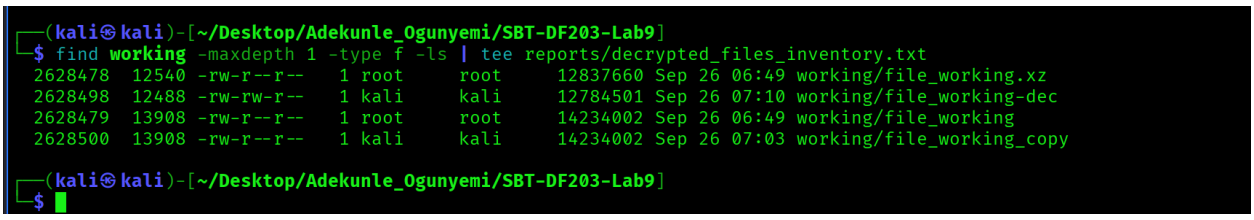


8.2 Decrypted File Inventory

find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt

Output:

- working/file_working.xz – 12,837,660 bytes
- working/file_working-dec – 12,784,501 bytes (decrypted capture)
- working/file_working – 14,234,002 bytes
- working/file_working_copy – 14,234,002 bytes



8.3 Hashes of Working Files

sha256sum working/* | tee reports/all_working_file_hashes.txt

Output:

c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working_copy
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz



9. Part E – Extract Endpoints and Conversations

9.1 Decrypted Capture Path

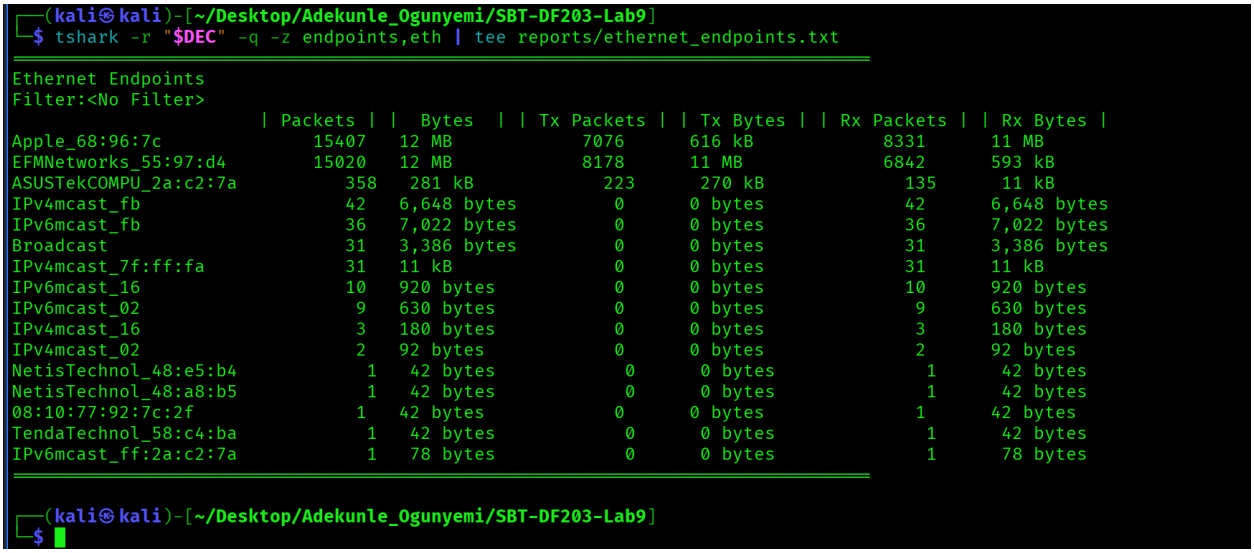
DEC=$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt

9.2 Ethernet Endpoints

tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

Key endpoints:

- Apple_68:96:7c – 154,071 packets / 12 MB
- EFMNetworks_55:97:dd – 150,201 packets / 12 MB
- ASUSTekCOMPU_2a:c2:7a – 35,828 packets / 1 MB

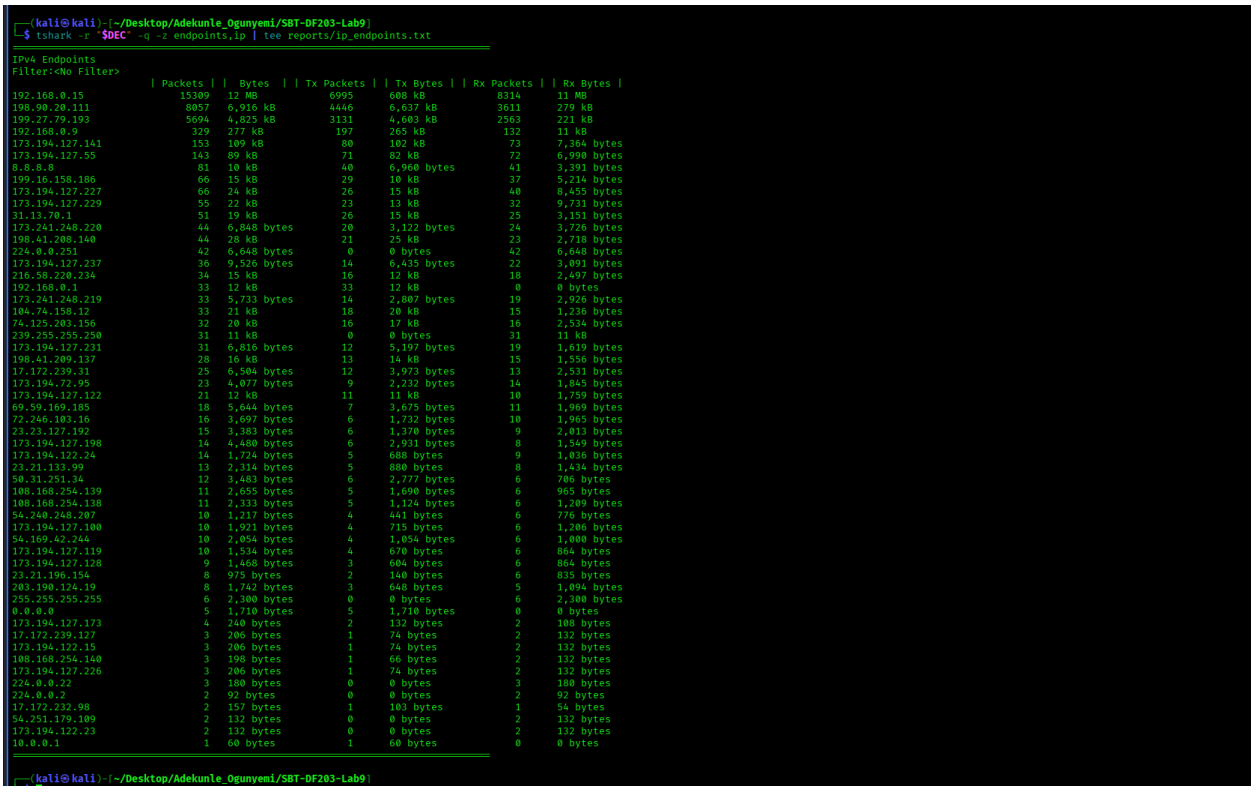


9.3 IP Endpoints

tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt

Key IP endpoints:

IP Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
192.168.0.15	15,309	12 MB	6,995	608 kB	8,314	11 MB
198.90.20.111	8,057	6,916 kB	4,446	6,637 kB	3,611	279 kB
199.27.79.193	5,694	4,825 kB	3,131	4,603 kB	2,563	221 kB
192.168.0.9	329	277 kB	197	265 kB	132	11 kB
173.194.127.141	153	109 kB	80	102 kB	73	7,364 bytes
8.8.8.8	81	10 kB	40	6,960 bytes	41	3,391 bytes

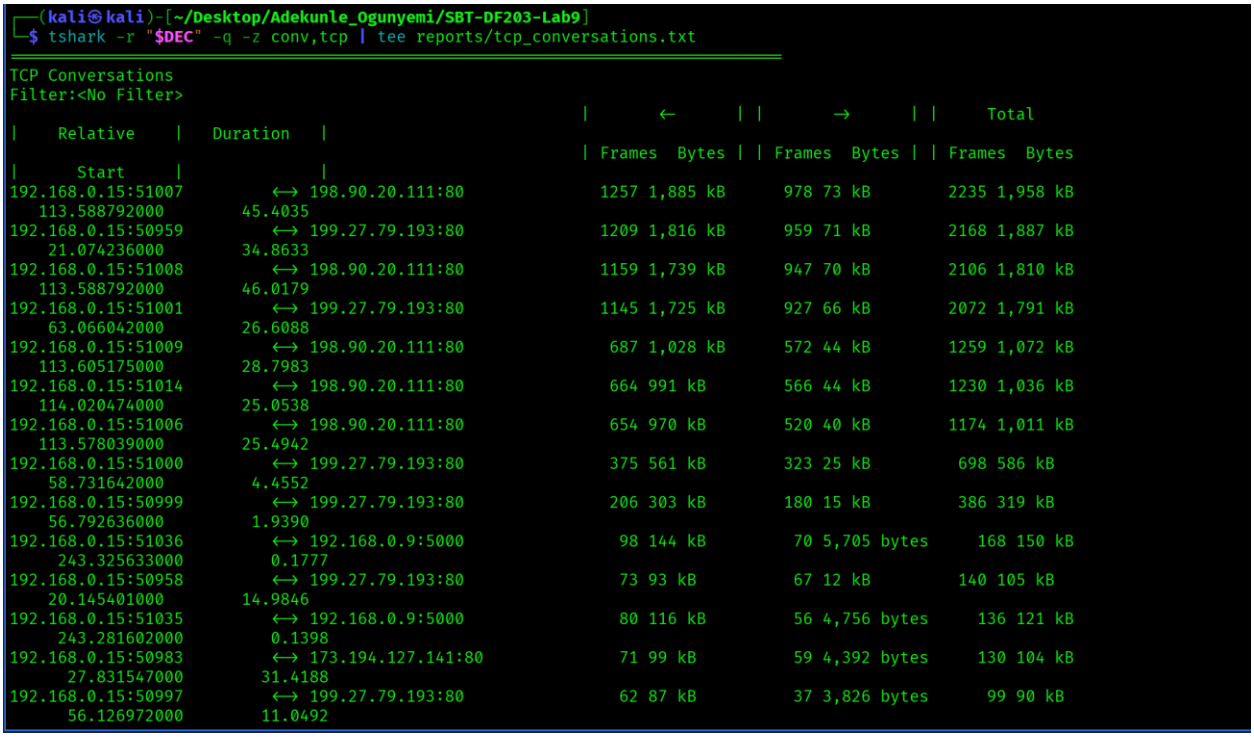


9.4 TCP Conversations

tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txttshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

Key conversations:

Conversation	Frames	Bytes	Duration
192.168.0.15:51007 ↔ 198.90.20.111:80	2,235	1,958 kB	45.4035 s
192.168.0.15:50959 ↔ 199.27.79.193:80	2,168	1,887 kB	34.8633 s
192.168.0.15:51008 ↔ 198.90.20.111:80	2,106	1,810 kB	46.0179 s
192.168.0.15:51001 ↔ 199.27.79.193:80	2,072	1,791 kB	26.6088 s
192.168.0.15:51009 ↔ 198.90.20.111:80	1,259	1,072 kB	28.7983 s



```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
================================================================================
TCP Conversations
Filter:<No Filter>
                                               |       ←      | |       →      | |     Total
|    Relative    |   Duration   |
                                               | Frames  Bytes | | Frames  Bytes | | Frames  Bytes
|      Start     |              |
192.168.0.15:51007         <-> 198.90.20.111:80              1257 1,885 kB      978 73 kB        2235 1,958 kB
   113.588792000        45.4035
192.168.0.15:50959         <-> 199.27.79.193:80              1209 1,816 kB      959 71 kB        2168 1,887 kB
    21.074236000        34.8633
192.168.0.15:51008         <-> 198.90.20.111:80              1159 1,739 kB      947 70 kB        2106 1,810 kB
   113.588792000        46.0179
192.168.0.15:51001         <-> 199.27.79.193:80              1145 1,725 kB      927 66 kB        2072 1,791 kB
    63.066042000        26.6088
192.168.0.15:51009         <-> 198.90.20.111:80               687 1,028 kB      572 44 kB        1259 1,072 kB
   113.605175000        28.7983
192.168.0.15:51014         <-> 198.90.20.111:80               664 991 kB        566 44 kB        1230 1,036 kB
   114.020474000        25.0538
192.168.0.15:51006         <-> 198.90.20.111:80               654 970 kB        520 40 kB        1174 1,011 kB
   113.578039000        25.4942
192.168.0.15:51000         <-> 199.27.79.193:80               375 561 kB        323 25 kB         698 586 kB
    58.731642000         4.4552
192.168.0.15:50999         <-> 199.27.79.193:80               206 303 kB        180 15 kB         386 319 kB
    56.792636000         1.9390
192.168.0.15:51036         <-> 192.168.0.9:5000                98 144 kB         70 5,705 bytes     168 150 kB
     243.325633000         0.1777
192.168.0.15:50958         <-> 199.27.79.193:80                73 93 kB          67 12 kB         140 105 kB
    20.145401000        14.9846
192.168.0.15:51035         <-> 192.168.0.9:5000                80 116 kB         56 4,756 bytes     136 121 kB
     243.281602000         0.1398
192.168.0.15:50983         <-> 173.194.127.141:80              71 99 kB          59 4,392 bytes     130 104 kB
      27.831547000        31.4188
192.168.0.15:50997         <-> 199.27.79.193:80                62 87 kB          37 3,826 bytes      99 90 kB
      56.126972000        11.0492
```

9.5 IP/MAC Mapping Sample

tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
| head -n 200 | tee reports/ip_mac_mapping_sample.tsv

The output showed:

- DHCP traffic between 0.0.0.0 and 255.255.255.255
- DNS queries from 192.168.0.15 to 8.8.8.8
- IGMPv2 traffic to 224.0.0.2 and 224.0.0.251
- TCP and TLSv1 traffic between 192.168.0.15 and 17.172.239.31

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ tshark -r "$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \
  | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7                       0.0.0.0 255.255.255.255 DHCP
9                       0.0.0.0 255.255.255.255 DHCP
11                      192.168.0.1     255.255.255.255 DHCP
20                      192.168.0.15    8.8.8.8 DNS
21                      192.168.0.15    8.8.8.8 DNS
22                      192.168.0.15    224.0.0.2       IGMPv2
23                      192.168.0.15    224.0.0.251     IGMPv2
24                      192.168.0.15    224.0.0.2       IGMPv2
25                      192.168.0.15    224.0.0.251     IGMPv2
26                      8.8.8.8 192.168.0.15    DNS
27                      8.8.8.8 192.168.0.15    DNS
30                      192.168.0.15    17.172.239.31   TCP
31                      192.168.0.15    17.172.239.127  TCP
33                      17.172.239.31   192.168.0.15    TCP
35                      192.168.0.15    17.172.239.31   TCP
36                      192.168.0.15    17.172.239.31   TLSv1
37                      17.172.239.127  192.168.0.15    TCP
38                      192.168.0.15    17.172.239.127  TCP
42                      17.172.239.31   192.168.0.15    TCP
43                      17.172.239.31   192.168.0.15    TLSv1
44                      17.172.239.31   192.168.0.15    TLSv1
45                      17.172.239.31   192.168.0.15    TLSv1
47                      0.0.0.0 255.255.255.255 DHCP
48                      0.0.0.0 255.255.255.255 DHCP
49                      192.168.0.15    17.172.239.31   TCP
50                      192.168.0.15    17.172.239.31   TCP
51                      192.168.0.1     192.168.0.15    DHCP
52                      192.168.0.15    17.172.239.31   TLSv1
55                      17.172.239.31   192.168.0.15    TCP
56                      192.168.0.15    17.172.239.31   TLSv1
57                      17.172.239.31   192.168.0.15    TCP
58                      17.172.239.31   192.168.0.15    TLSv1
```

9.6 ARP and DHCP Evidence

The decrypted capture also revealed ARP requests and DHCP transactions:

0.000000 EFMNetworks_55:97:d4 > Broadcast ARP 42 Who has 192.168.0.15? Tell 192.168.0.1
0.944186 Apple_68:96:7c > NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
0.944185 0.0.0.0 > 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
0.954435 192.168.0.1 > 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6be1c
1.030202 192.168.0.15 > 8.8.8.8 DNS 73 Standard query 0x9583 A www.apple.com

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
    1   0.000000 EFMNetworks_55:97:d4 → Broadcast    ARP 42 Who has 192.168.0.15? Tell 192.168.0.1
    2   0.943674 Apple_68:96:7c → NetisTechnol_48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.168.1.41
    3   0.944186 Apple_68:96:7c → NetisTechnol_48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
    4   0.944186 Apple_68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? Tell 192.168.1.12
    5   0.944186 Apple_68:96:7c → EFMNetworks_55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
    6   0.944186 Apple_68:96:7c → TendaTechnol_58:c4:ba ARP 42 Who has 192.168.10.1? Tell 192.168.10.113
    7   0.944185      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e6be1c
    8   0.944704 EFMNetworks_55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
    9   0.947778      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request  - Transaction ID 0×21e6be1c
   10   0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2      ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
   11   0.954435  192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK      - Transaction ID 0×21e6be1c
   12   0.954937 Apple_68:96:7c → Broadcast    ARP 42 ARP Announcement for 192.168.0.15
   13   0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2      ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
   14   0.956994 Apple_68:96:7c → Broadcast    ARP 42 ARP Announcement for 192.168.0.15
   15   0.959034 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   16   0.960066 Apple_68:96:7c → Broadcast    ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
   17   1.026106 Apple_68:96:7c → Broadcast    ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
   18   1.029698 Apple_68:96:7c → Broadcast    ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
   19   1.029696 EFMNetworks_55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
   20   1.030202 192.168.0.15 → 8.8.8.8      DNS 73 Standard query 0×9583 A www.apple.com
   21   1.186368 192.168.0.15 → 8.8.8.8      DNS 85 Standard query 0×be40 A 48-courier.push.apple.com
   22   1.187392 192.168.0.15 → 224.0.0.2    IGMPv2 46 Leave Group 224.0.0.251
   23   1.187392 192.168.0.15 → 224.0.0.251  IGMPv2 46 Membership Report group 224.0.0.251
   24   1.187904 192.168.0.15 → 224.0.0.2    IGMPv2 46 Leave Group 224.0.0.251
   25   1.189439 192.168.0.15 → 224.0.0.251  IGMPv2 46 Membership Report group 224.0.0.251
   26   1.350208      8.8.8.8 → 192.168.0.15 DNS 203 Standard query response 0×9583 A www.apple.com CNAME www.
```

10. Part F – Extract Images, HTML and Other Objects

10.1 HTTP Object Export

mkdir -p exported/http_objects exported/foremost
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 |
tee reports/http_export_log.txt

10.3 Exported File Types

find exported -type f -exec file {} \; | tee
reports/exported_file_types.txt

Recovered files included:

- GIF images: 00001260.gif, 00010876.gif, 00003719.gif, 00003653.gif, 00000406.gif, 00002450.gif, 00010456.gif
- HTML documents: 00001639.htm, 00024427.htm
- PNG images: 00020736.png (19x19), 00001831.png (12x12), 00001765.png (44x44), 00010214.png (1x1), 00024242.png (19x19), 00010671.png (1x400)
- JPEG images: 00004538.jpg (718x404), 00000104.jpg (2492x2492)
- Text file: audit.txt
- HTTP objects: of_destiny.png, bootstrap.min.css, accordion.min.js, trackalyze.js, dynamik-min.css, lumbar.gif

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
exported/foremost/gif/00001260.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010876.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003719.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00003653.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00000406.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00002450.gif: GIF image data, version 89a, 1 x 1
exported/foremost/gif/00010456.gif: GIF image data, version 89a, 1 x 1
exported/foremost/htm/00001639.htm: HTML document, ASCII text
exported/foremost/htm/00024427.htm: HTML document, ASCII text
exported/foremost/png/00020736.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00001831.png: PNG image data, 12 x 12, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00001765.png: PNG image data, 44 x 44, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00010214.png: PNG image data, 1 x 1, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00024242.png: PNG image data, 19 x 19, 8-bit/color RGBA, non-interlaced
exported/foremost/png/00010671.png: PNG image data, 1 x 400, 8-bit/color RGBA, non-interlaced
exported/foremost/jpg/00004538.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72×72, segme
nt length 16, comment: "Lavc55.69.100", baseline, precision 8, 718×404, components 3
exported/foremost/jpg/00000104.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1×1, segment len
gth 16, baseline, precision 8, 2492×2492, components 3
exported/foremost/audit.txt: ASCII text
exported/http_objects/YDAoLskQQ5MOAgvHUQCcLQa6gm6bS00u2Qn-iPLo1Go.woff: Web Open Font Format, TrueType, length
22824, version 1.1
exported/http_objects/of_destiny.png%3fv=h%2Bw3qHw8Nrlcs96j8ZN%2BqruehW91KTg3Y9jHy9f3jgs243%2FhnENQoVopWm1m2Z2l
H6T7uGA4V5E%3D: PNG image data, 1 x 1, 8-bit/color RGBA, non-interlaced
exported/http_objects/bootstrap.min.css: ASCII text, with very long lines (65307)
exported/http_objects/Zd2E9abXLFGSr9G3YK2MsNxB8OB85xaNTJvVSB9YUjQ.woff: Web Open Font Format, TrueType, length
13224, version 1.1
```

10.4 Exported File Hashes

find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt

Example hashes:

- 8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 – exported/foremost/gif/00001260.gif
- a0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8 – exported/foremost/gif/00000406.gif
- 3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6 – exported/foremost/jpg/00004538.jpg
- e97ffa450c3e5b483fd008453b6170f97dbec5c6cf096c8223a6b332c861376b – exported/http_objects/YDAoLskQQ5MOAgvHUQCcLQa6gm6bS00u2Qn-iPLo1Go.woff

```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF203-Lab9]
└─$ find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00001260.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00010876.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003719.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015  exported/foremost/gif/00003653.gif
a0d3a0aff7dc3bf32d2176fc3dcda6e7aba2867c4f4d1f7af6355d2cfc6c44f8  exported/foremost/gif/00000406.gif
ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629  exported/foremost/gif/00002450.gif
53f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94  exported/foremost/gif/00010456.gif
1d11ac950ff1be55ca0e69a487331563f958afd8295252ca88c96c4e72b161eb  exported/foremost/htm/00001639.htm
02e357aea1b284a9e6a71088574431019174 0a96d99bb189146b1cbc547ea79e  exported/foremost/htm/00024427.htm
792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676  exported/foremost/png/00020736.png
534c09b9b2d436a30c260eaad82679d2a2c970d4962399a8da227ee07162a755  exported/foremost/png/00001831.png
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d  exported/foremost/png/00001765.png
ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a  exported/foremost/png/00010214.png
681fb2c21414bfb0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf  exported/foremost/png/00024242.png
ddf5dd4e0ef2b185e8bb0af7b6e90ebe74a84384cb4700658e76e754c8bfe550  exported/foremost/png/00010671.png
f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc  exported/foremost/jpg/00004538.jpg
3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6  exported/foremost/jpg/00000104.jpg
0c46db27cce55d5bd7080d696748d74f89ae10e39c29bf3a6d244c694a195a55  exported/foremost/audit.txt
e97ffa450c3e5b483fd008453b6170f97dbec5c6cf096c8223a6b332c861376b  exported/http_objects/YDAoLskQQ5MOAgvHUQCcLQa
6gm6bS00u2Qn-iPLo1Go.woff
ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a  exported/http_objects/of_destiny.png%3fv=h%2B
w3qHw8Nrlcs96j8ZN%2BqruehW91KTg3Y9jHy9f3jgs243%2FhnENQoVopWm1m2Z2lH6T7uGA4V5E%3D
186c40d06fc13830497a7b9f42bc14538c1b7fa0b98560c7911e6e1a935f769d  exported/http_objects/bootstrap.min.css
5eafcd482cbd31c604c630b54ce55bfc87ce483f25a9b91a8c2c2fd4c4557d15  exported/http_objects/Zd2E9abXLFGSr9G3YK2MsNx
B8OB85xaNTJvVSB9YUjQ.woff
055e154927bba6b4a204c7142fd3945a4006bccb9b526e76e59f8dafebc0405b  exported/http_objects/accordion.min.js
de90760c360ef82aa247fdc2b79ce6cc0e465b119d70aec68e89d7850fe07569  exported/http_objects/object6524
59ebace7202b39be46546a909c5ae92c2a34245cba32537b63564c22169e34cf  exported/http_objects/trackalyze.js
fe4529ba03075f4ba177e0898b91def18bad106bb55a1ef4a2560bf982ceaff8  exported/http_objects/dynamik-min.css
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b  exported/http_objects/lumbar.gif%3fa=%7B%22UI
```

11. Part G – Optional Historical Passphrase Challenge

This part was optional and instructor-assessed. The slide discussed deriving a seed and cracking the CTF passphrase using supplied challenge information. The 40-bit WEP key A4:3D:F6:F3:74 is different from a human-readable router passphrase. The known key, derived seed, character constraints, and challenge hash were recorded. Only the instructor-provided fndseed.py or crackWEP40 script would be used against the historical dataset. No live networks were targeted.

12. WEP Security Analysis

1. WEP is cryptographically unsafe for several reasons:
2. Short 24-bit IV: The IV space is only 2^24, so IVs repeat quickly on busy networks.
3. RC4 keystream reuse: When the same IV is used, the same RC4 keystream is reused, leaking relationships between plaintexts.

4. Statistical key recovery: Repeated IVs and RC4 biases allow statistical attacks that recover the key.
5. Weak integrity: WEP uses CRC-32 for integrity, which is not cryptographically secure and can be forged.
6. No replay protection: WEP does not prevent replay attacks.
7. No protected management frames: Management frames are not protected.

The recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported. Legacy WEP should be removed rather than strengthened with longer passphrases.

13. Required Findings Worksheet

Question	**Finding**
Capture format and duration	pcap, IEEE 802.11, 272.97 seconds
BSSID	00:26:66:55:97:d6
Primary station MAC addresses	00:26:66:55:97:d4, 04:8d:38:48:8e:b4, 08:10:77:92:7c:f2, c8:3a:55:97:d6
Protected frame count	15,477 WEP data packets
Repeated IV evidence	Many IVs repeated; repeated IVs confirmed WEP weakness
Recovered/validated WEP40 key	A4:3D:F6:F3:74
Decrypted capture filename/hash	working/file_working-dec – 167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04
Host IP and MAC	192.168.0.15 – Apple_68:96:7c
Server IP and MAC	198.90.20.111 – external HTTP server
Key protocols observed	ARP, DHCP, DNS, TCP, TLSv1, IGMPv2
Recovered images/HTML	GIF, PNG, JPEG, HTML, CSS, JS, WOFF
Optional router passphrase	Protected appendix only
Security conclusion	WEP is broken; migrate to WPA2-AES/CCMP or WPA3

14. Required Screenshots Checklist

Screenshot	Description
1	Compressed evidence file and hash
2	Decompressed capture identification
3	Protocol hierarchy
4	802.11 frame categories
5	Protected WEP frames and IV evidence
6	Aircrack-ng result
7	Airdecap-ng decryption result
8	Decrypted IP/MAC endpoint inventory
9	Exported image or HTML objects
10	Hashes of decrypted capture and recovered files

15. Academic Integrity and Authorisation Statement

I confirm that I completed this lab in an authorised environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.

Signed: Adekunle Ogunyemi
Date: 25 September 2026